1. *Explain the difference between an event-driven application and a console-based application.*
   - **Event-driven Application**: A program that responds to actions from the user, like clicking a button.
   - **Console-based application**: A program that uses text for the user to enter information and receive output.
2. *Explain how code is executed in an event-driven application.*
   - In an event-driven application the code waits for an event, like a button click, and then runs the code connected to that event.
3. *Can components be added directly to a frame? Explain*.
   - Components are individual items in a GUI that a user can interact with, like a button, text box, or label. They are usually added to a panel not the frame directly however some components can be added directly to a frame. Adding components to a panel helps with organization, layout, and appearance therefore making it easier to manage.
4. *Can a label respond to events? Explain.*
   - A label can technically respond to events, such as a mouse click, if an event listener is added to it. However, labels are usually used to display text or images instead of interacting with the user.
5. *Why do you think a GUI needs to be run from an event-dispatching thread?*
   - A GUI needs to run an event-dispatching thread so that user actions, like clicking buttons, are handled properly and the interface stays responsive and updates correctly.
6. *What is the difference between a label and a button?*
   - A **Label** displays text or an image, while a **Button** can be clicked by the user to perform an action.

8. List three ways to control the layout of a content pane.
   - **FlowLayout** - Arranges components in a row.
   - **BorderLayout** - Arranges components into five areas.
   - **GridLayout** - Arranges components in rows and columns.

13. What must first be done with numeric data typed in a text field before it can be used in a calculation?
   - The numeric data must be converted from a String to a number (such as an int**,** or **double**) before it can be used in a calculation.

14. What is the value of num1 in the last statement below?

```
double num1;
Double num2 = new Double(3);
String num3 = "5"; num1 = num2.doubleValue() + Double.valueOf(num3).doubleValue();
```

Value of num1 is 8.0

15. *An application prompts a user to select a name from a list of six names. Which is a better component choice: a text field or a combo box? Explain.*
   - A combo bow is the better choice, it allows the user to select a name from a list of six options instead of typing it manually, limiting time and mistakes.